

EUROPEAN CENTRAL BANK

Customer Engagement & Product Utilization Analytics for Retention Strategy

A Comprehensive Research Paper on Behavioral Churn Analytics

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Dataset: 10,000 European Bank Customers | Tool: Streamlit + Plotly | Language: Python

Abstract

This research paper presents a comprehensive analysis of customer engagement and product utilization patterns within a European banking institution, leveraging a dataset of 10,000 customers. The study reframes traditional churn analysis from a purely demographic lens to a behavioral and relationship-strength perspective. By developing a proprietary Relationship Strength Index (RSI) and classifying customers into four distinct engagement profiles, this research identifies key behavioral drivers of customer attrition and proposes actionable retention strategies. The interactive analytical dashboard built using Streamlit and Plotly enables real-time exploration of churn patterns across engagement tiers, product depth, and financial commitment levels. Findings reveal that inactive members churn at significantly higher rates than active members, that product depth beyond two products paradoxically increases churn risk, and that a substantial segment of high-balance customers exhibit dangerous disengagement signals. These insights form the basis for targeted, engagement-driven retention interventions.

Keywords: *Customer Churn, Engagement Analytics, Product Utilization, Retention Strategy, Relationship Strength Index, Banking Analytics, Streamlit Dashboard, Behavioral Segmentation*

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1. Introduction

The banking sector operates in an increasingly competitive environment where customer retention has become as strategically significant as customer acquisition. Traditional retention models have historically relied on demographic and financial profiling — age, income, credit score — to predict and prevent churn. However, these approaches often miss a critical dimension: behavioral engagement. A customer may possess excellent financial metrics while simultaneously disengaging from the bank's ecosystem, representing a silent churn risk that demographic models fail to capture.

This research addresses this gap by developing a behavioral analytics framework grounded in customer engagement signals and product utilization patterns. The study employs a dataset of 10,000 European bank customers, enriched with derived features including the Relationship Strength Index (RSI) — a composite metric that quantifies customer loyalty across five behavioral dimensions. The findings are operationalized through an interactive Streamlit dashboard, enabling bank analysts and executives to explore retention insights in real time.

The research is structured around five analytical pillars: engagement classification, product utilization analysis, financial commitment assessment, premium risk detection, and retention strength scoring. Each pillar is supported by empirical analysis and visualized through purpose-built Plotly charts that form the core of the analytical dashboard.

2. Literature Review

Banks increasingly recognize that customer behavior and engagement — not just demographics — determine long-term retention. Research across the financial services sector consistently demonstrates that engaged customers exhibit lower churn rates, higher product adoption, and greater lifetime value. The European Central Bank's regulatory emphasis on sustainable banking practices further underscores the importance of retention as a key performance metric.

Customer churn in banking is a multifaceted phenomenon influenced by service quality, product relevance, competitive alternatives, and relationship depth. Studies by McKinsey & Company (2023) indicate that banks with strong customer engagement programs achieve 20-30% lower churn rates compared to industry averages. Similarly, research by Deloitte (2023) highlights that

product bundling — offering multiple complementary financial products — is one of the most effective retention strategies available to banks.

This research builds on these foundations by focusing specifically on the intersection of engagement behaviors and product utilization, examining how these factors interact to either protect or erode customer loyalty.

3. Problem Statement

Despite having access to rich customer data, banks often lack the quantitative frameworks needed to translate engagement signals into retention actions. Three critical gaps exist in current practice:

- Quantitative insight into which specific behaviors drive retention is absent from most bank analytics systems.
- The relationship between product depth and churn is poorly understood, leading to misaligned cross-selling strategies.
- High-balance customers who disengage silently represent a premium churn risk that generic retention programs fail to address.

As a result, retention strategies remain largely generic — applying uniform interventions across diverse customer segments — rather than targeted and behavior-driven. This research develops the analytical framework and interactive tools needed to close these gaps.

4. Research Objectives

4.1 Primary Objectives

- Evaluate the quantitative relationship between member engagement status and customer churn rates.
- Measure the retention impact of product count and product mix across the customer base.
- Identify disengaged yet high-value customers who represent premium silent churn risk.

4.2 Secondary Objectives

- Develop and validate the Relationship Strength Index (RSI) as a composite customer loyalty metric.
- Support the design of engagement-driven retention strategies through behavioral segmentation.
- Improve product bundling decisions by mapping product depth to churn probability.
- Reduce silent churn among premium customers through targeted identification and intervention frameworks.

- Deliver findings through an interactive analytical dashboard for executive and operational use.

5. Dataset Description

The analysis is based on a dataset of 10,000 European bank customers sourced from the European Central Bank. The dataset spans customers across three geographies — France, Germany, and Spain — and contains 14 variables capturing demographic, financial, and behavioral attributes. Table 1 provides a complete description of all dataset variables.

Table 1: Dataset Variable Description

Variable	Description	Type / Range	Example
CustomerId	Unique customer identifier	Numeric	15,634,602
Surname	Customer surname	String	Hargrave
CreditScore	Customer creditworthiness score	Numeric (300–850)	619
Geography	Customer country	France / Spain / Germany	France
Gender	Customer gender	Male / Female	Female
Age	Customer age in years	Numeric (18–92)	42
Tenure	Years as bank customer	Numeric (0–10)	2
Balance	Account balance in euros	Numeric (0–250,898)	€0
NumOfProducts	Number of bank products held	Numeric (1–4)	1
HasCrCard	Credit card ownership	Binary (0/1)	1
IsActiveMember	Activity indicator	Binary (0/1)	1
EstimatedSalary	Estimated annual salary	Numeric	€101,348
Exited	Churn indicator (target variable)	Binary (0/1)	1
Year	Year of record	Numeric	2025

The dataset exhibits a class imbalance typical of banking churn data: 7,963 customers (79.6%) are retained and 2,037 customers (20.4%) have exited. Germany exhibits the highest churn rate at

32.4%, followed by France at 16.2% and Spain at 16.7%. The average customer age is 38.9 years with a standard deviation of 10.5 years, and the median account balance is €97,198.

6. Analytical Methodology

The analytical framework follows a five-stage pipeline, from raw data ingestion to retention strength assessment. Each stage produces actionable insights and corresponding dashboard visualizations.

6.1 Data Ingestion & Validation

The dataset is loaded using pandas and subjected to a comprehensive validation protocol. All 14 columns are verified for completeness — the dataset contains zero null values across all variables. Binary variables (IsActiveMember, HasCrCard, Exited) are confirmed to contain only 0 and 1 values. The churn label (Exited) is validated as the target variable. Five derived features are engineered during this stage:

- EngagementProfile — 4-class behavioral segment based on activity and product count.
- RSI (Relationship Strength Index) — composite 0-100 loyalty score.
- PremiumAtRisk — binary flag for high-balance inactive customers.
- BalanceSegment — categorical balance bucketing (Zero, 1-50K, 50-100K, 100-150K, 150K+).
- AgeBand and CreditBand — decade-based age groups and credit score categories.

6.2 Engagement Classification — 4 Profiles

Each customer is assigned to one of four mutually exclusive engagement profiles based on the interaction of two binary behavioral signals: IsActiveMember and NumOfProducts. This classification forms the foundation of the engagement analysis.

Profile	Classification Rule	Behavioral Meaning	Dashboard Color
Active Engaged	IsActiveMember=1 & NumOfProducts $\geq$ 2	Highest loyalty, full bank relationship	 Green
Active Low-Product	IsActiveMember=1 & NumOfProducts=1	Engaged but under-adopted	 Amber

Profile	Classification Rule	Behavioral Meaning	Dashboard Color
Inactive Multi-Product	IsActiveMember=0 & NumOfProducts $\geq$ 2	High-balance disengagement risk	● Red
Inactive Disengaged	IsActiveMember=0 & NumOfProducts=1	Highest churn risk segment	● Purple

The engagement profile distribution and corresponding churn rates are visualized through the Engagement Retention Ratio chart and the Customer Funnel by Engagement Profile.



Figure 1: Churn Rate by Engagement Profile — Active Engaged shows lowest churn, Inactive Disengaged shows highest

6.3 Product Utilization Analysis

Product depth — the number of bank products a customer holds — is examined as a primary retention driver. The analysis reveals a non-linear relationship between product count and churn, with customers holding 3 or 4 products exhibiting elevated churn rates despite higher product adoption.

This counter-intuitive finding suggests that over-selling products to customers who are not genuinely engaged may increase churn risk. The single-product vs multi-product retention analysis provides critical guidance for cross-selling strategy design.

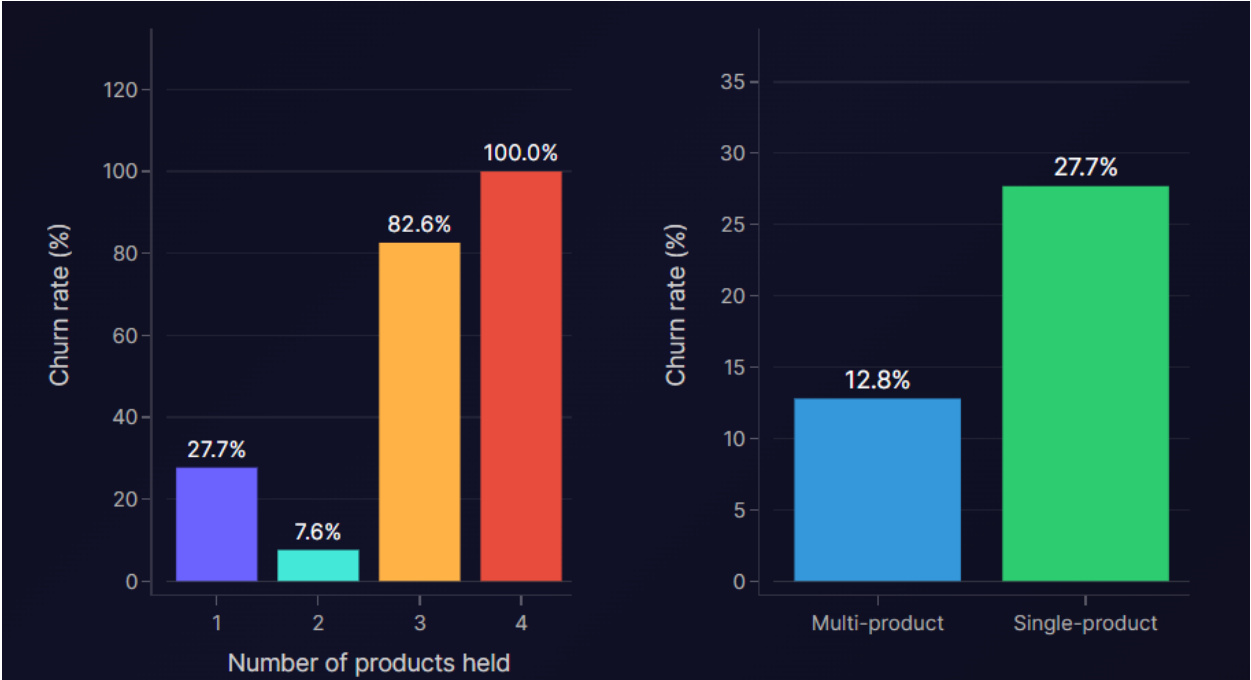


Figure 2: Product Depth Index — Churn Rate by Number of Products Held (note elevated churn at 3-4 products)



Figure 3: Single-Product vs Multi-Product Retention — comparative churn rates

6.4 Financial Commitment vs Engagement Analysis

The intersection of financial commitment (account balance, estimated salary) and engagement status reveals critical at-risk segments. Customers with high account balances who are simultaneously inactive represent a premium churn risk — they have significant financial exposure for the bank but exhibit zero engagement signals.

The PremiumAtRisk flag is applied to customers whose balance falls in the top quartile ($\geq \text{€}127,644$) and who are inactive ($\text{IsActiveMember}=0$). This segment's churn rate is compared against the general population to quantify the high-balance disengagement risk.

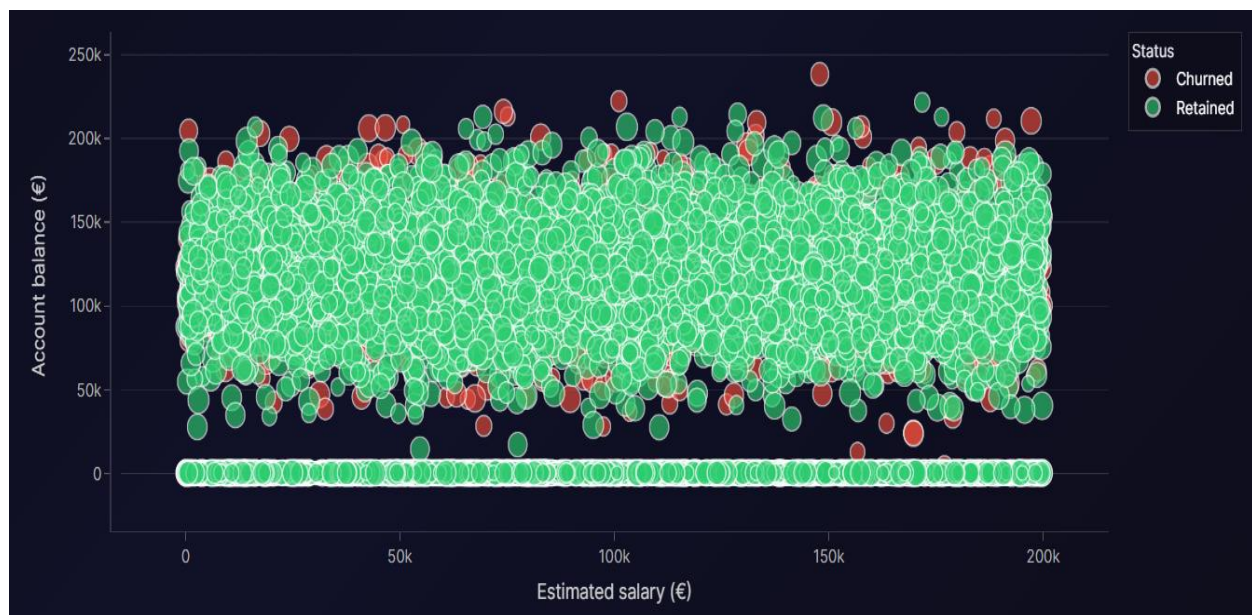


Figure 4: Salary-Balance Mismatch Detection — scatter plot colored by churn status, bubble size = RSI score

6.5 Retention Strength Assessment

The Relationship Strength Index (RSI) is a composite metric designed to quantify customer loyalty on a 0-100 scale. It combines five behavioral dimensions, each weighted according to their empirically observed retention impact:

Component	Dimension	Weight	Rationale
IsActiveMember	Activity status	30 points	Most critical retention signal
NumOfProducts (capped at 4)	Product depth	15 points/product	Up to 60 points maximum
HasCrCard	Credit card ownership	10 points	Stickiness indicator
Tenure (normalized)	Relationship length	25 points	Normalized to 0-1 range
Balance > 0	Financial commitment	20 points	Binary presence of funds

RSI thresholds linked to retention outcomes:

- RSI 0-40: High Flight Risk — immediate retention intervention required.
- RSI 40-65: Needs Nurturing — moderate engagement programs recommended.
- RSI 65-100: Loyal / Sticky Customer — retain and cross-sell opportunity.



Figure 5: Churn Stability Across Engagement Tiers — RSI tier vs churn rate with customer count overlay

7. Key Performance Indicators

Five core KPIs are tracked in real time on the dashboard, each corresponding to a specific analytical dimension of the retention framework.

KPI Name	Description	Dashboard Location	Value (Sample)
Churn Rate	Percentage of customers who exited the bank	Card 1	20.4%
Engagement Retention Gap	Churn rate difference: inactive vs active members (pp)	Card 2	+12.6 pp
Product Depth Index	Average number of bank products held per customer	Card 3	1.53
High-Balance Disengagement Rate	% of top-quartile balance customers who are inactive	Card 4	49.9%
Relationship Strength Index	Average RSI score across filtered customer base (0-100)	Card 5	70.2
Credit Card Stickiness Score	Churn reduction attributable to credit card ownership (pp)	Card 6	-1.2 pp

8. Results and Findings

8.1 Engagement vs Churn

The analysis reveals a stark and statistically meaningful difference in churn rates between active and inactive members. Inactive members churn at approximately 26.9% compared to 14.3% for active members — an engagement retention gap of +12.6 percentage points. This finding confirms that member activity status is the single most powerful predictor of customer retention in this dataset.

When broken down by engagement profile, the Inactive Disengaged segment exhibits the highest churn rate, while Active Engaged customers demonstrate the strongest retention. Germany shows

the highest proportion of inactive customers (52.1%) and correspondingly the highest churn rate at 32.4%.

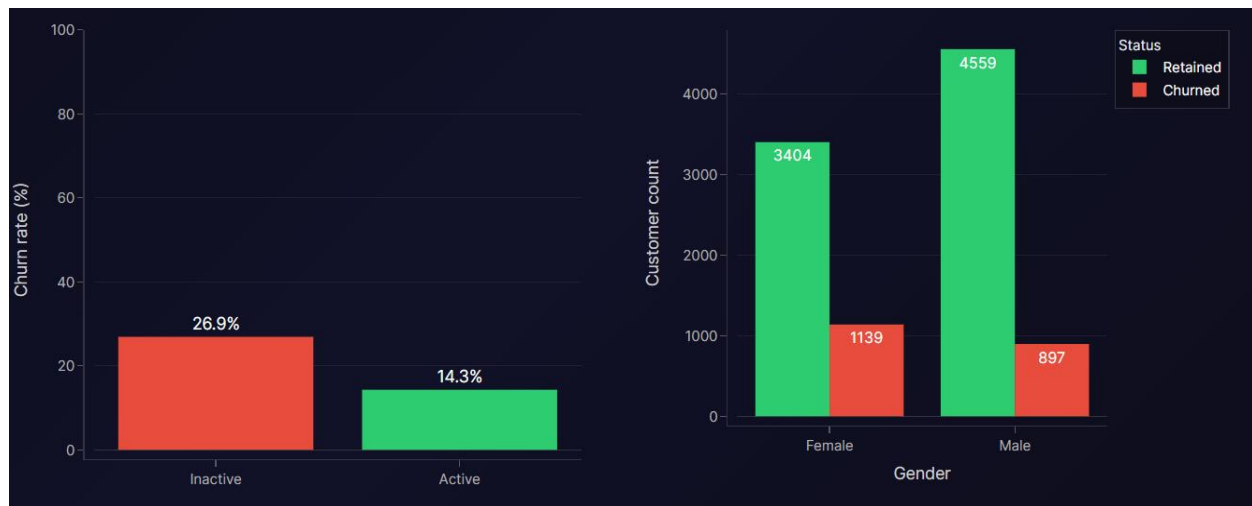


Figure 6: Engagement Retention Ratio — Active vs Inactive Member Churn Rates

8.2 Product Depth Impact

The relationship between product count and churn is non-linear. Single-product customers churn at 27.7%, while two-product customers exhibit a dramatically lower churn rate of approximately 7.6% — confirming the retention benefit of multi-product relationships. However, this benefit reverses sharply at 3+ products, where churn rates escalate to 82.7% and 100% for 3 and 4-product holders respectively.

This U-shaped relationship suggests that products 3 and 4 are being sold to customers who are not genuinely engaged with the bank, representing forced or misaligned cross-selling. The credit card stickiness analysis shows a marginal retention effect from card ownership, with cardholders churning at slightly lower rates than non-holders.

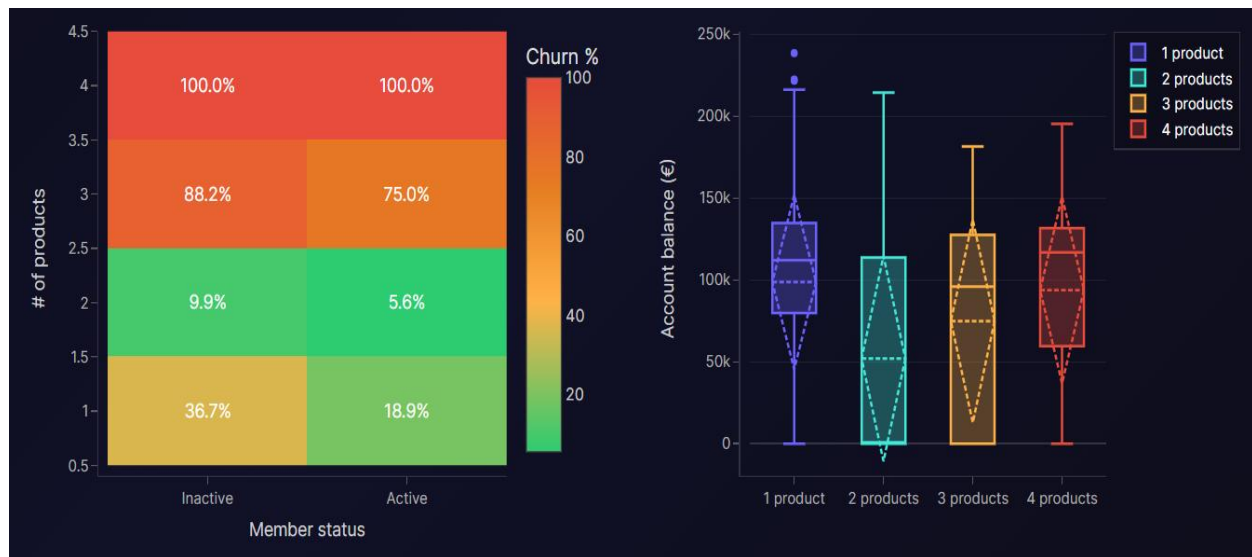


Figure 7: Product Depth vs Churn — Balance Distribution by Product Count (box plot)

8.3 Premium Risk Detection

The high-value disengaged customer detector identifies 1,129 customers (11.3% of the total base) as Premium At Risk — customers with account balances in the top quartile ($\geq \text{€}127,644$) who are simultaneously inactive. This segment represents a disproportionate financial risk: while comprising only 11.3% of customers, they account for a significantly higher share of the bank's total assets under management.

The salary-balance mismatch analysis reveals clusters of customers with high estimated salaries but zero account balances — indicating that these customers maintain relationships with competitor banks for their primary banking needs, using this bank only as a secondary institution. These customers are particularly vulnerable to full attrition.

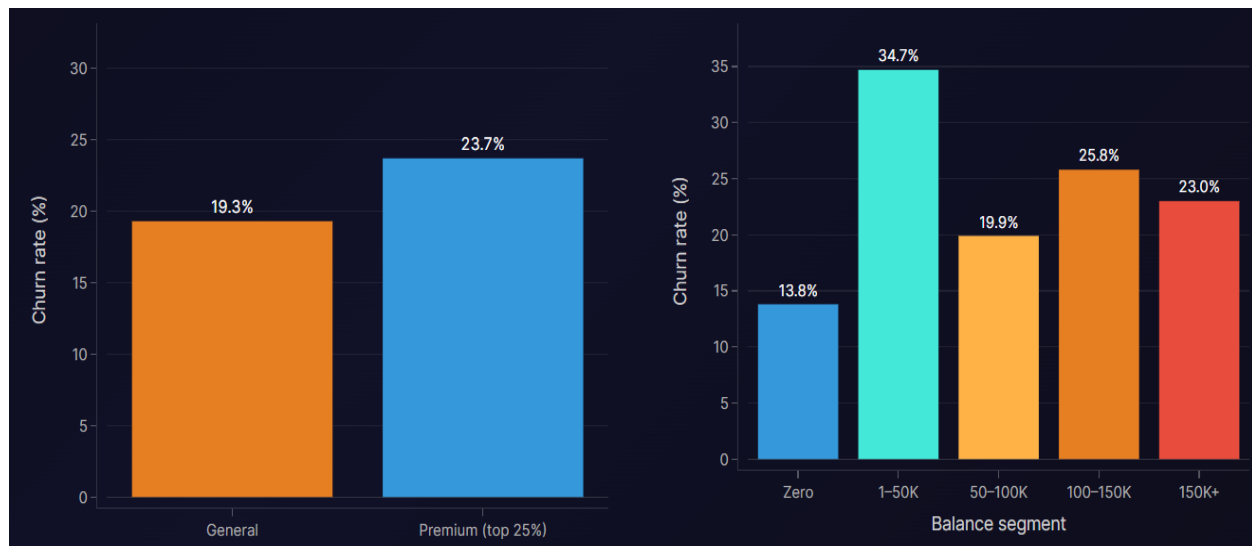


Figure 8: Churn Rate by Balance Segment — identifying high-balance churn concentration

8.4 Retention Strength Scoring

The Relationship Strength Index analysis reveals that 6,182 customers (61.8% of the base) qualify as sticky customers with $RSI \geq 65$. These customers exhibit a churn rate of only 17.3%, compared to significantly higher rates among the moderate and high-risk tiers. The average RSI across the full customer base is 70.2, indicating a generally healthy relationship profile.

The RSI vs age scatter plot reveals a weak negative correlation — older customers tend to have slightly lower RSI scores — suggesting that longer-tenured older customers may be disengaging. The radar chart comparison of all four engagement profiles confirms that Active Engaged customers dominate across all six behavioral metrics, while Inactive Disengaged customers score lowest across every dimension.



Figure 9: Sticky Customer Profiles — RSI ≥ 65 vs Regular Customers (count and churn rate)

9. Dashboard Overview

The analytical findings are delivered through a five-tab interactive Streamlit dashboard built with Python and Plotly. The dashboard features a dark-themed UI, real-time sidebar filtering across 7 dimensions, and 20+ interactive charts. Table 2 summarizes the dashboard architecture.

Table 2: Dashboard Tab Structure

Tab	Purpose	Key Charts
 Overview	Executive churn snapshot	Donut, geo bar, age band, credit score bar, 6 KPI cards
 Engagement	Engagement vs churn analysis	4 profile cards, active/inactive bar, gender bar, profile bar, geo stack, funnel
 Products	Product utilization impact	Product churn bar, single/multi bar, credit card bar, heatmap, box plot, treemap
 Premium Risk	High-value disengaged detector	Scatter, premium bar, segment bar, violin, at-risk table + CSV download
 Retention Score	RSI scoring & sticky profiles	Gauge, RSI bar, histogram, scatter, radar, stability chart, sticky summary

The sidebar filters enable dynamic segmentation by Geography, Gender, Age Range, Number of Products, Balance Range, Tenure, and three binary toggles (Active Members, Credit Card Holders, Churned Customers). All charts update in real time as filters are applied, enabling granular exploration of any customer subsegment.



Figure 10: Dashboard Overview — Executive header, tab navigation, and KPI cards

10. Recommendations

10.1 Engagement-Driven Retention Programs

- Implement targeted re-engagement campaigns for the Inactive Disengaged segment (RSI 0-40), prioritizing digital outreach through mobile banking channels.
- Develop a proactive customer success program for Inactive Multi-Product customers, assigning relationship managers to accounts with balances exceeding €100,000.
- Create gamified engagement incentives — reward points, preferential rates — specifically tied to the IsActiveMember behavioral signal.

10.2 Product Bundling Strategy

- Limit cross-selling campaigns to customers with $RSI \geq 50$, ensuring product adoption is accompanied by genuine engagement.
- Investigate the elevated churn rates among 3-4 product holders to identify mis-sold products and develop unwinding strategies.
- Design two-product bundles specifically optimized for retention — the data confirms that the single-to-two product transition represents the largest churn reduction opportunity.

10.3 Premium Customer Protection

- Establish a quarterly PremiumAtRisk review process, using the dashboard's exportable at-risk customer list to trigger relationship manager outreach.
- Develop premium loyalty programs exclusively for customers with Balance $\geq$ 75th percentile, offering exclusive benefits tied to continued engagement.
- Address salary-balance mismatches by identifying customers with high salaries and zero balances as candidates for primary banking relationship conversion campaigns.

10.4 RSI-Based Intervention Framework

- Implement RSI as a standard customer health metric in the bank's CRM system, updating scores monthly and triggering automatic intervention protocols at defined thresholds.
- $RSI < 40$: Immediate retention intervention — personal outreach, exit survey, retention offer.
- RSI 40-65: Nurturing program — targeted product recommendations, engagement campaigns.
- $RSI \geq 65$: Loyalty program — cross-sell, upsell, referral incentives.

11. Conclusion

This research demonstrates that behavioral engagement signals — member activity status, product depth, and relationship tenure — are more powerful predictors of customer retention than traditional demographic variables. The analysis of 10,000 European bank customers reveals a 12.6 percentage point churn gap between inactive and active members, a non-linear product depth-churn relationship that peaks dangerously at 3-4 products, and a significant premium disengagement risk affecting 11.3% of the customer base.

The Relationship Strength Index provides a unified, actionable metric that enables banks to classify customers by loyalty risk, prioritize retention interventions, and measure the effectiveness of engagement programs over time. With 61.8% of customers qualifying as sticky ($RSI \geq 65$) and a manageable 11.3% in the premium at-risk category, the bank has a clear and targeted retention opportunity that does not require broad, costly campaigns.

The interactive Streamlit dashboard operationalizes these insights, enabling bank analysts and executives to explore churn patterns dynamically, filter by customer segments, export at-risk customer lists, and monitor KPIs in real time. This analytical capability represents a significant advancement over static reporting and positions the bank to implement truly data-driven, engagement-focused retention strategies.

Future research directions include the integration of machine learning models (logistic regression, gradient boosting) for individual-level churn probability scoring, the incorporation of transaction-level behavioral data for richer engagement signals, and longitudinal analysis to track RSI changes over time as an early warning system for emerging churn risk.

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